

# Signals and Systems

## Chapter 2

### Time-Domain Representations of Linear Time-Invariant Systems

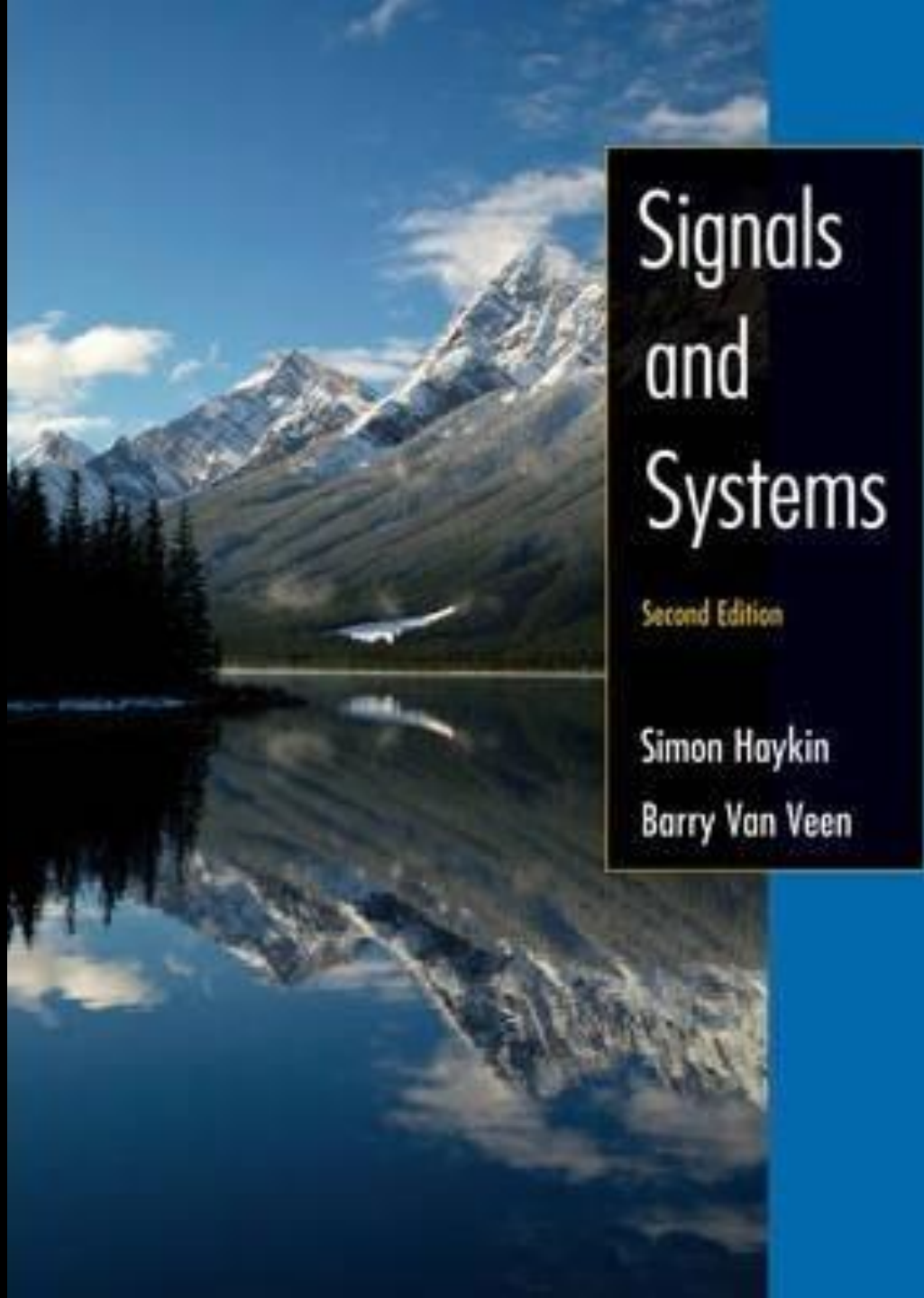
## Convolution

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*Kaynaklar :*

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Signals  
and  
Systems

Second Edition

Simon Haykin  
Barry Van Veen

# Chapter 2

## Time-Domain Representations of Linear Time-Invariant Systems

### Objectives of this chapter

1. Introduction
2. The Convolution Sum
3. Convolution Sum Evaluation Procedure
4. The Convolution Integral
5. Convolution Integral Evaluation Procedure
6. Interconnections of LTI Systems
7. Relations between LTI System Properties and the Impulse Response
8. Step Response
9. Differential and Difference Equation Representations of LTI Systems
10. Solving Differential and Difference Equations
11. Characteristics of Systems Described by Differential and Difference Equations
12. Block Diagram Representations
13. State-Variable Descriptions of LTI Systems
14. Exploring Concepts with MATLAB
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## 2. Time-Domain Representations of Linear Time-Invariant Systems

### 1. Introduction

In this chapter, We examine several methods for describing the relationship between the input and output signals of linear time invariant (LTI) system

Firstly,

- Methods for describing the relationship between the input and output signals (as functions of time) of linear time-invariant (LTI) systems in time domain
- Analyzing and predicting the behavior of LTI systems
  - Both continuous-time and discrete-time
- characterizing LTI systems as its impulse response,
  - output of system due to unit impulse input applied at  $t=0$  or  $n = 0$
- Impulse response completely characterizes behavior of any LTI system

## 2. Time-Domain Representations of Linear Time-Invariant Systems

- Impulse response of a discrete-time system is obtained by setting the input equal to the impulse  $\delta[n]$
- In continuous-time case, the impulse signal is approximated by a pulse of large amplitude and brief duration
- By linearity and time invariance, the output signal must be a weighted superposition of time-shifted impulse responses
- This weighted superposition is termed the
  - convolution sum for discrete-time systems and
  - convolution integral for continuous-time systems

## 2. Time-Domain Representations of Linear Time-Invariant Systems

- The second method we shall examine for characterizing the input-output behavior of LTI systems is the linear constant-coefficient differential equation (for continuous-time ) or difference equation (for discrete-time systems)
- The third method of system representation we discuss is the system as an interconnection of block diagrams as three elementary operations:
  - scalar multiplication, addition and either a time shift for discrete-time systems or integration for continuous-time systems
- The final time-domain system representation discussed in this chapter is the state variable description
  - a series of coupled first-order differential or difference equations that represent the behavior of the system's "state" and
  - an equation that relates that state to output of the system
  - The state is a set of variables associated with energy storage or memory devices in the system

## 2.2 The Convolution Sum

- In discrete-time case, an arbitrary signal is expressed as a weighted superposition of shifted impulses
- Then, the convolution sum is obtained by applying a signal represented in this manner to an LTI system
- A similar procedure is for continuous-time systems
- Let a signal  $x[n]$  be multiplied by impulse sequence  $\delta[n]$ ; i.e.

$$x[n] \delta[n] = x[0] \delta[n]$$

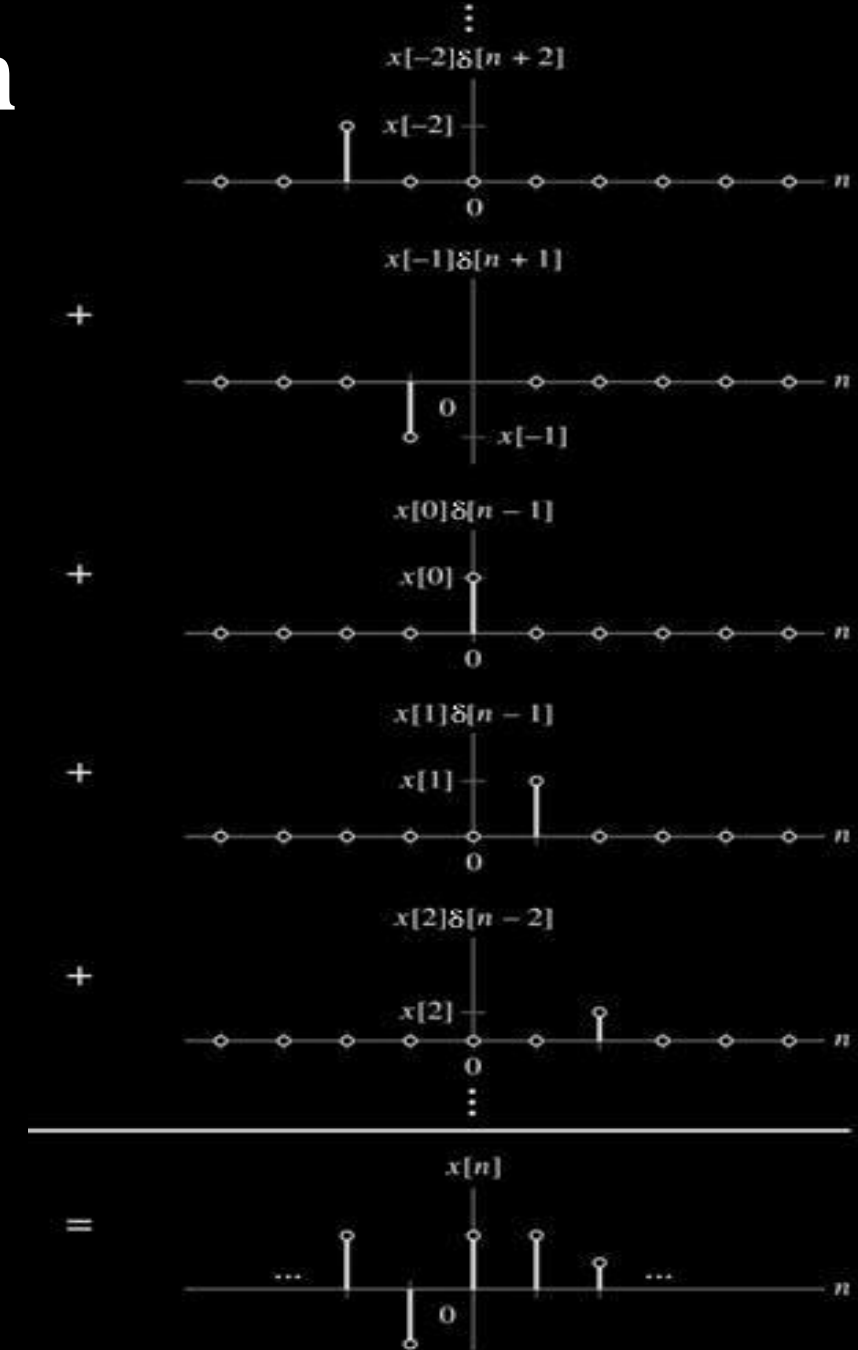
- This relationship may be generalized to product of  $x[n]$  and a time-shifted impulse sequence, to obtain

$$x[n] \delta[n-k] = x[k] \delta[n-k]$$

- Where  $n$  represents the time index,  $x[n]$  the entire signal,  $x[k]$  a specific value of the signal  $x[n]$  at time  $k$ .

- Lets represent  $x[n]$  as weighted sum of time-shifted impulse

$$x[n] = \dots + x[-2] \delta[n+2] + x[-1] \delta[n+1] + x[0] \delta[n] + x[1] \delta[n-1] + x[2] \delta[n-2] + \dots$$



**Figure 2.1 (p. 99)** Graphical example illustrating the representation of a signal  $x[n]$  as a weighted sum of time-shifted impulses.

- Which can be re-written as

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] \quad (2.1)$$

- Which is graphically illustrated in Fig .2.1
- When  $x[n]$  is passed through a system denoted by H, output is

$$y[n] = H\{x[n]\} = H\left\{ \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] \right\}$$

- Can also be written as

$$y[n] = \sum_{k=-\infty}^{\infty} H(x[k] \delta[n - k]) = \sum_{k=-\infty}^{\infty} x[k] H(\delta[n - k]) \quad (2.2)$$

- Defining

$$H(\delta[n - k]) = h[n - k] \quad (2.3)$$

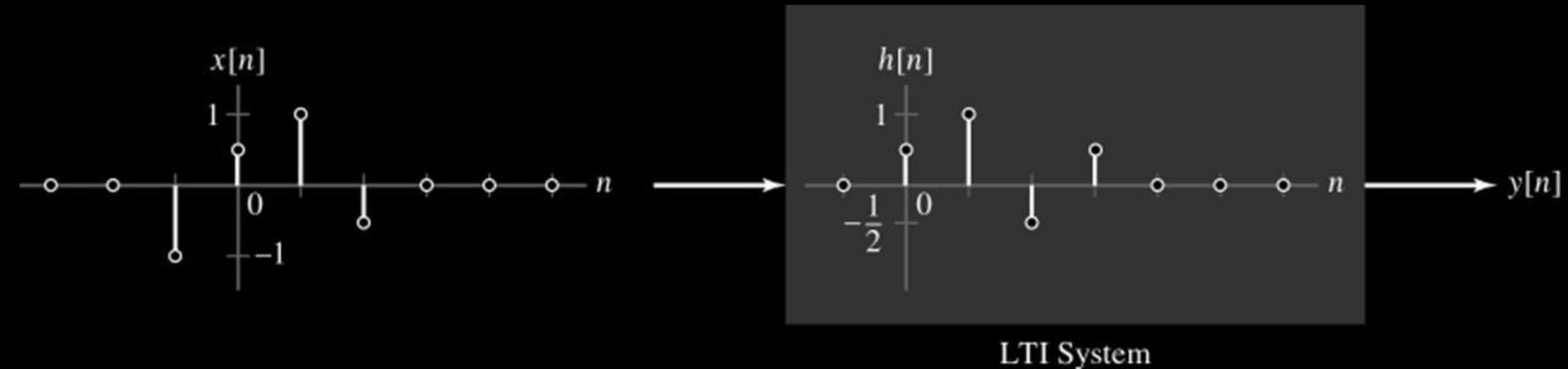
$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

- Which is the convolution process illustrated in Fig. 2.2.

**Figure 2.2a (p. 100)**

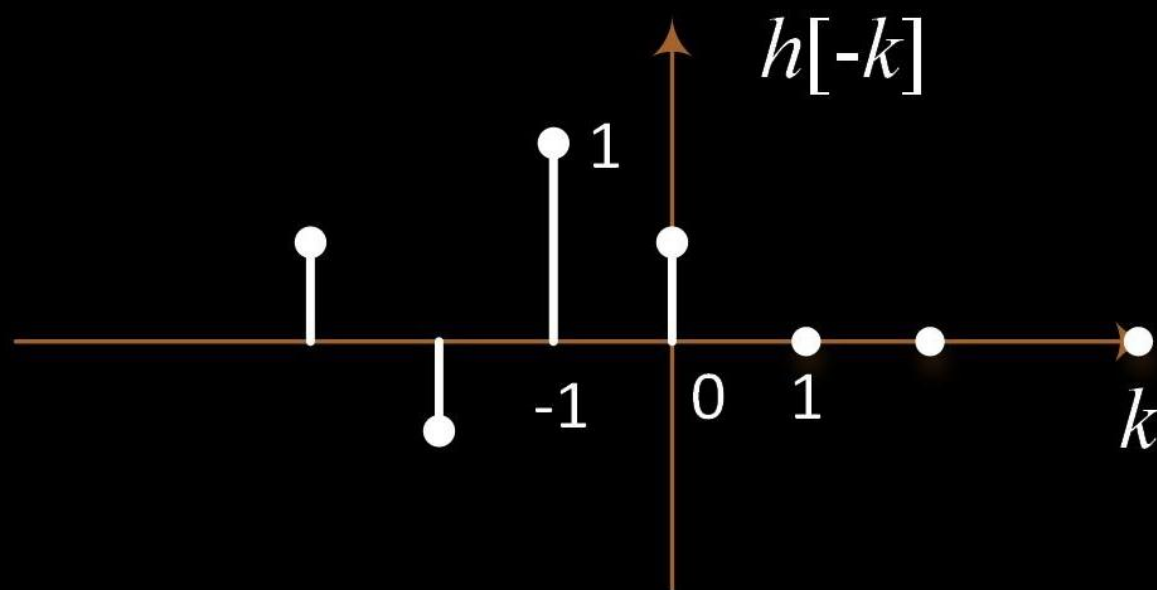
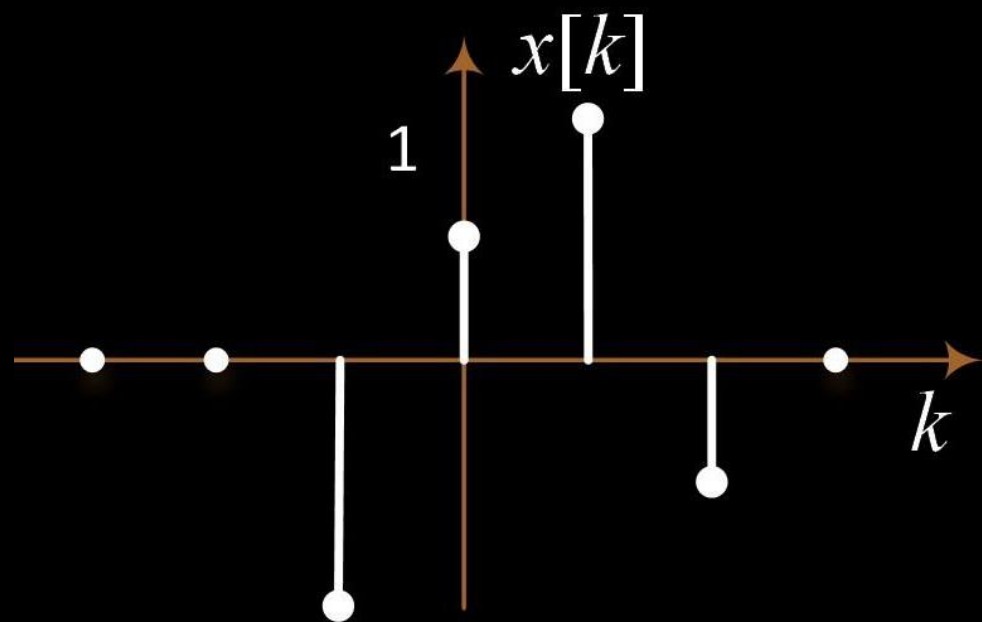
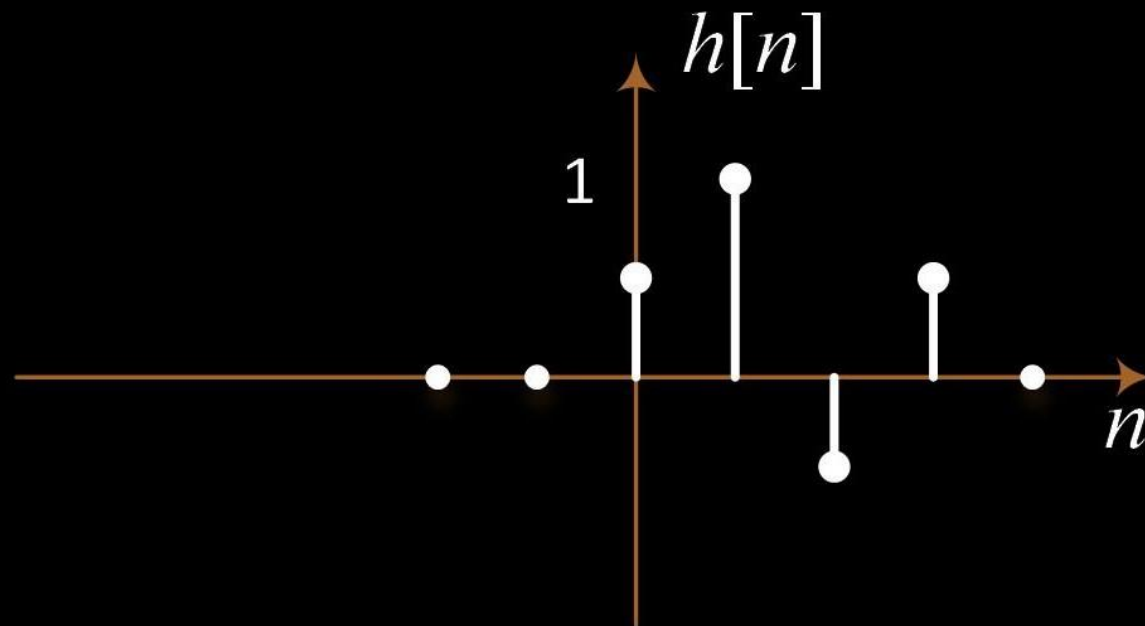
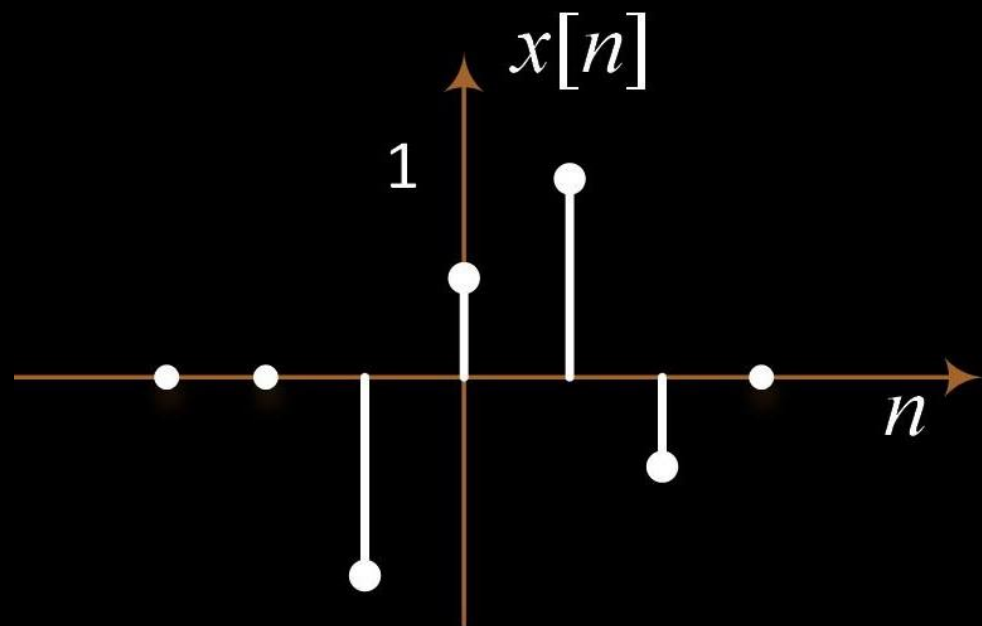
Illustration of the convolution sum. (a) LTI system with impulse response  $h[n]$  and input  $x[n]$ .

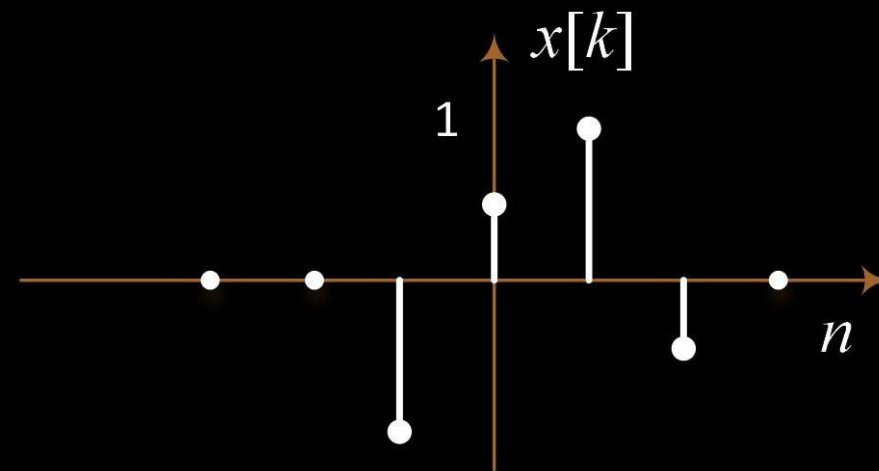
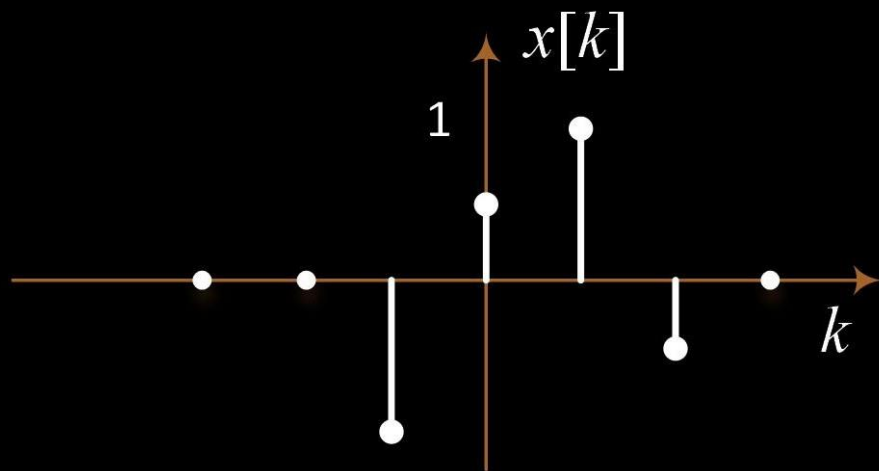
$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$



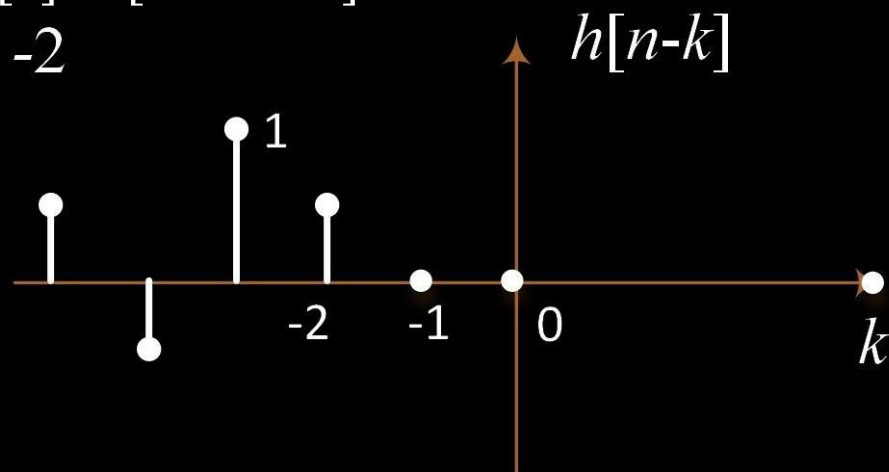
(a)



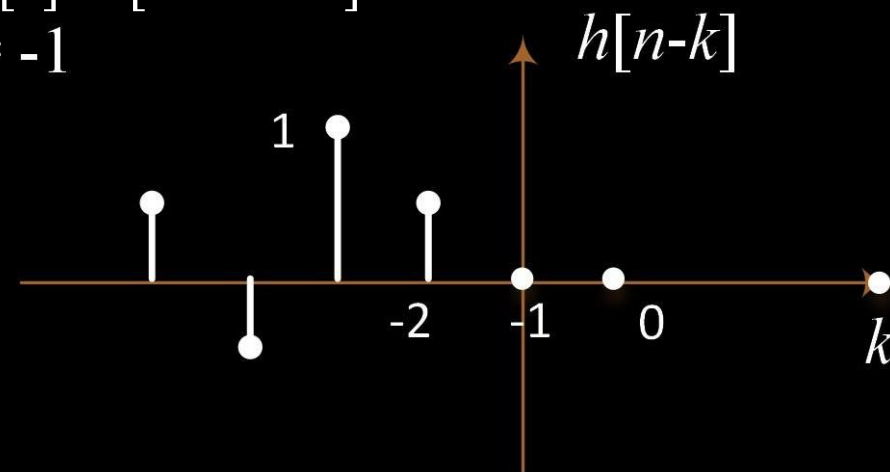




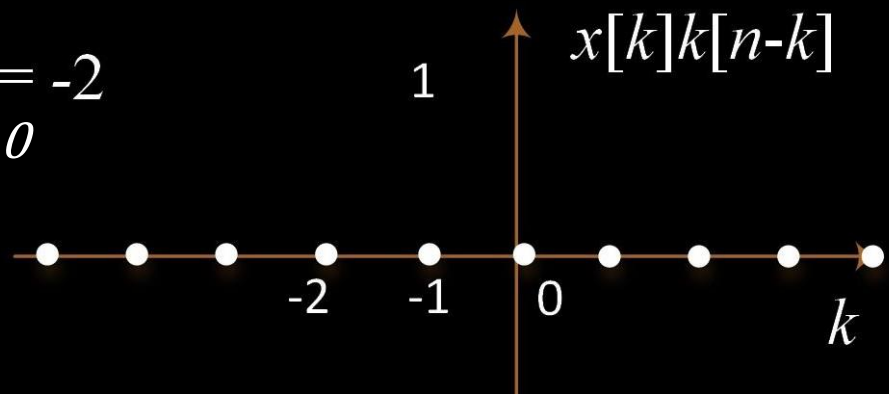
$y[-2] = \sum_k x[k] h[-2-k]$   
For  $n = -2$



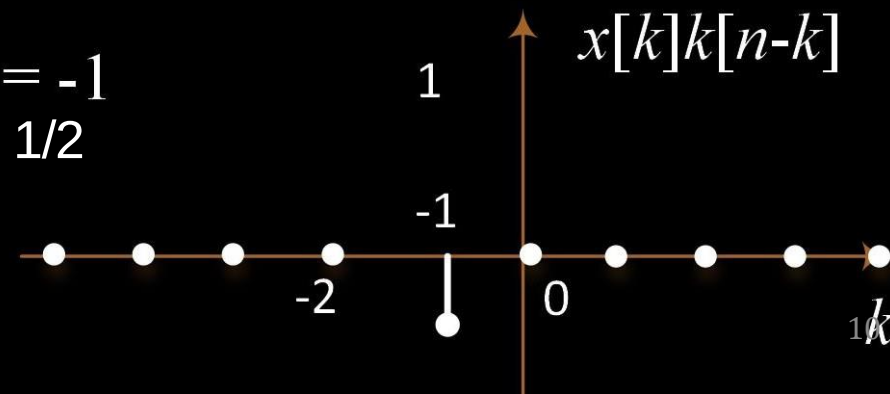
$y[-1] = \sum_k x[k] h[-1-k]$   
For  $n = -1$

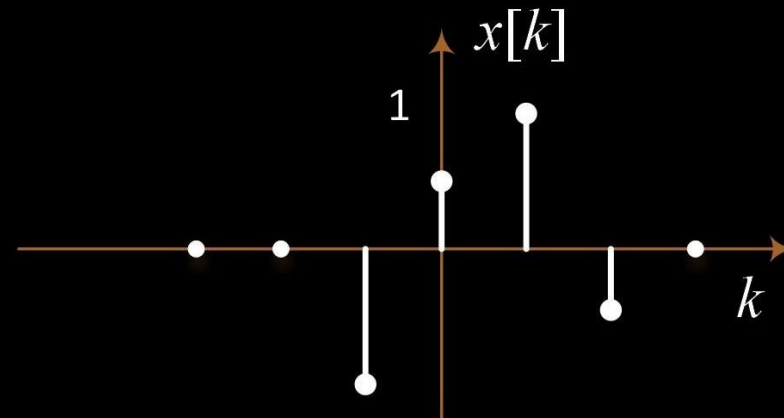
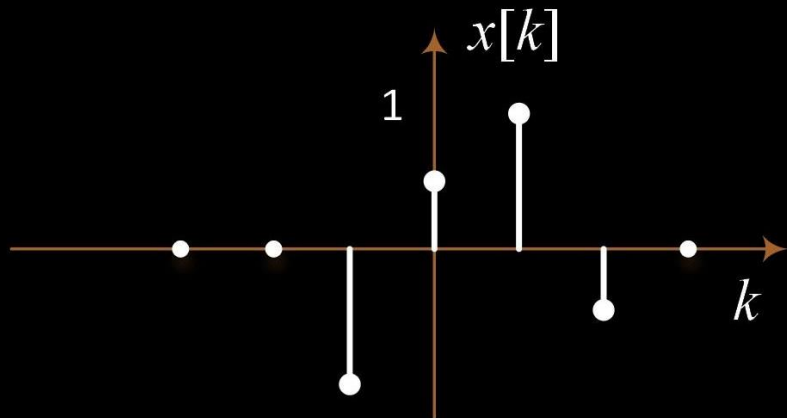


For  $n = -2$   
 $y[-2] = 0$

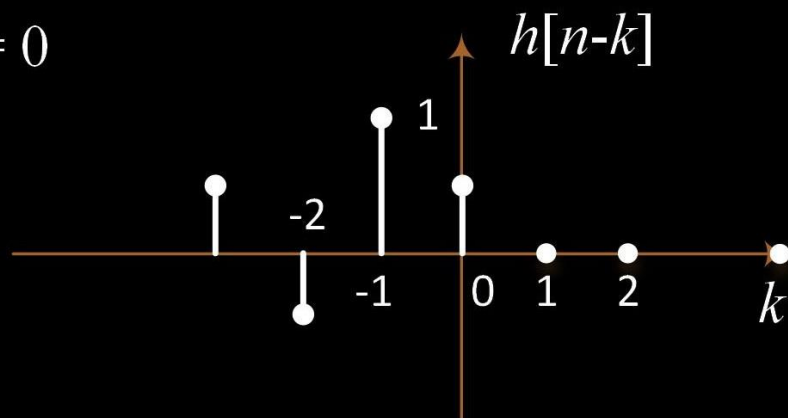


For  $n = -1$   
 $y[-1] = 1/2$

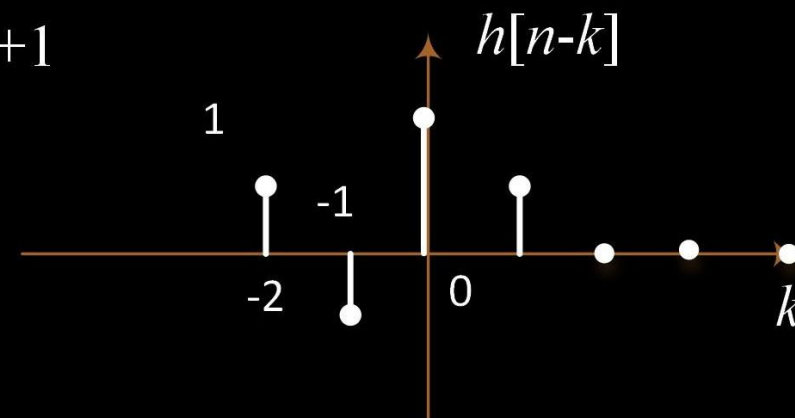




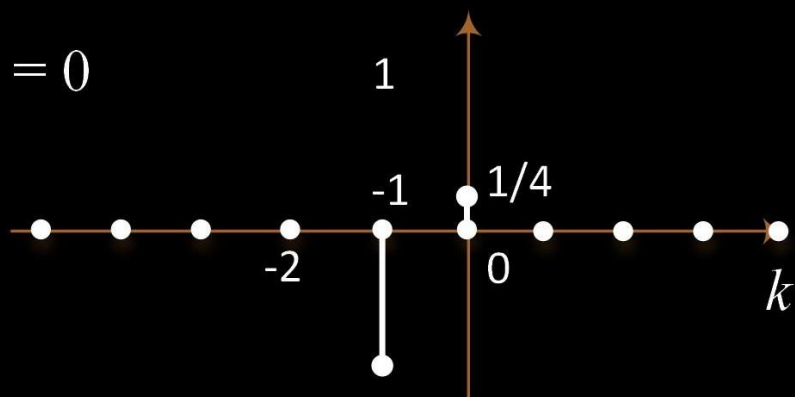
For  $n = 0$



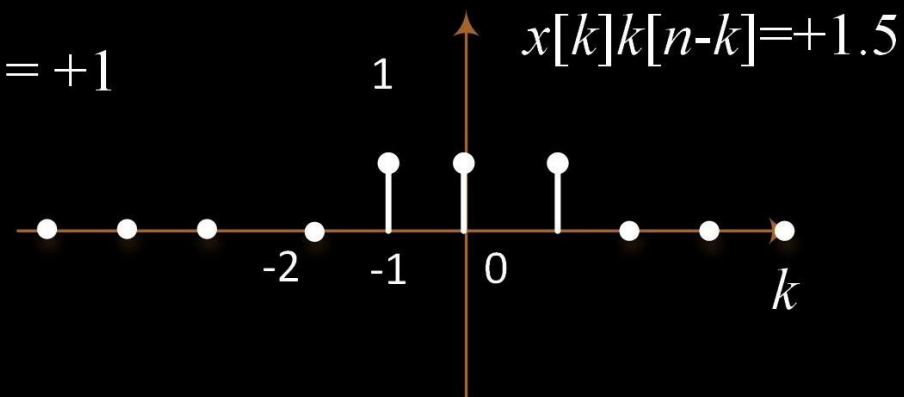
For  $n = +1$



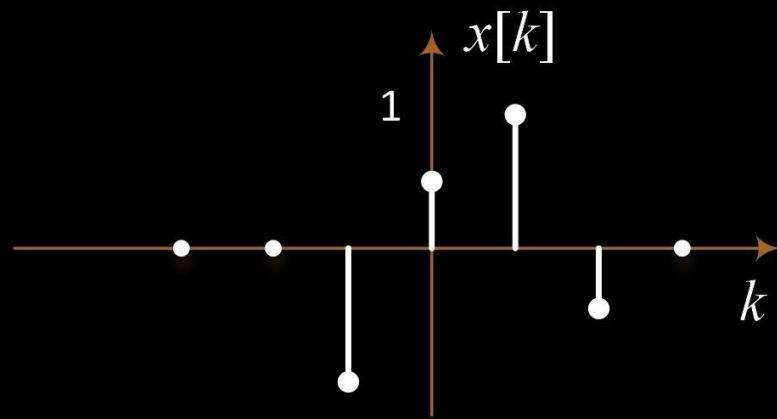
For  $n = 0$



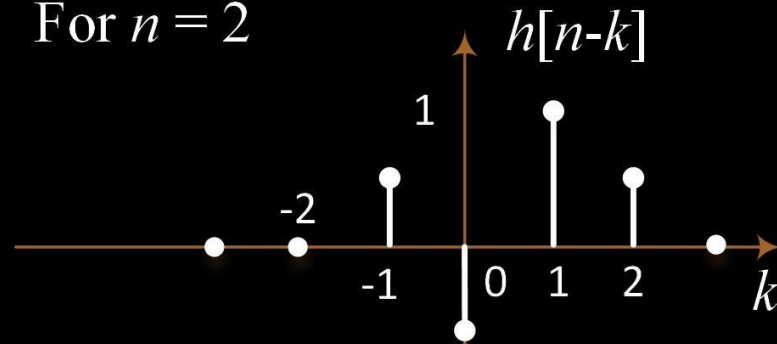
For  $n = +1$



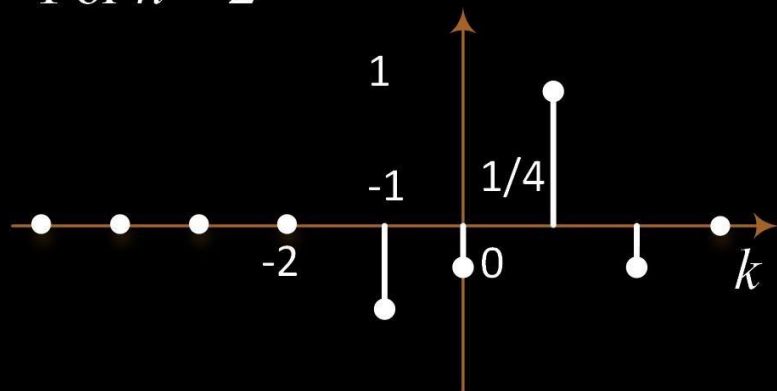
$$x[k]h[n-k] = -1 + 1/4 = -3/4$$



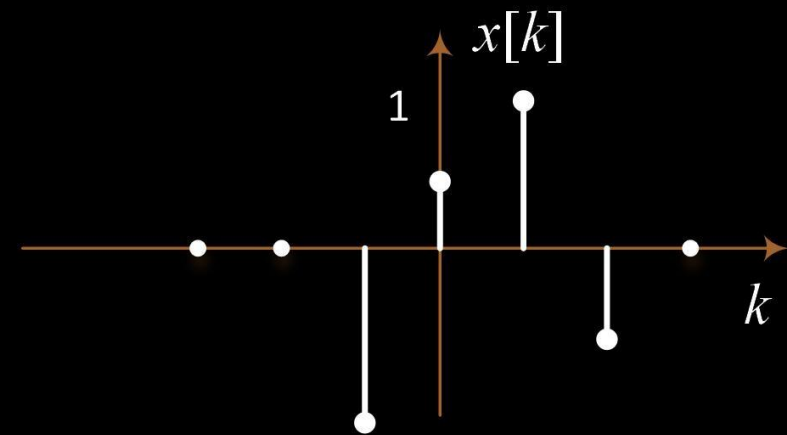
For  $n = 2$



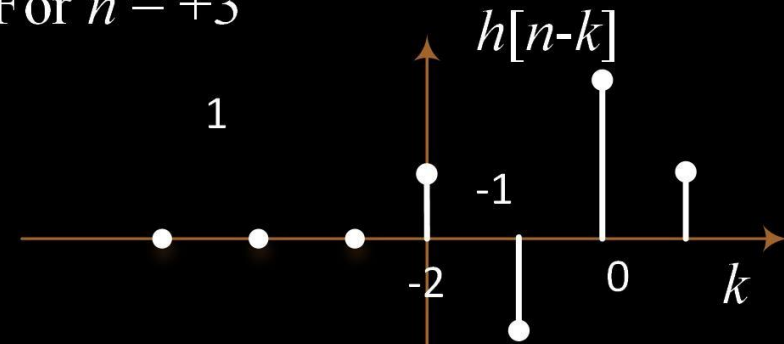
For  $n = 2$



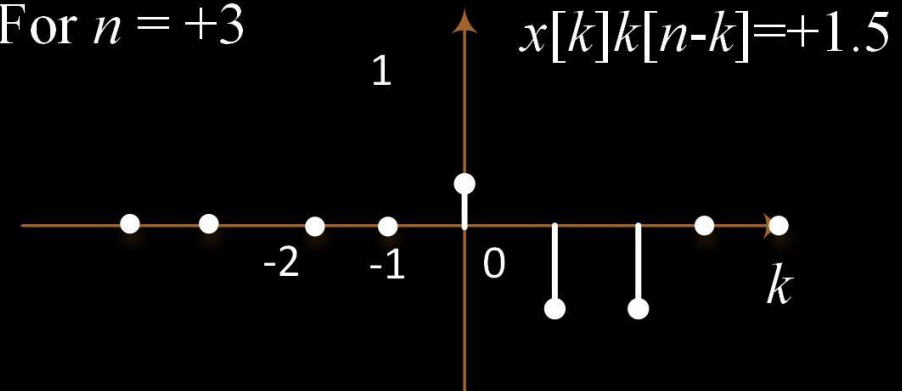
$$x[k]h[n-k] = -\frac{1}{2} - \frac{1}{4} + 1 - \frac{1}{4} = 0$$



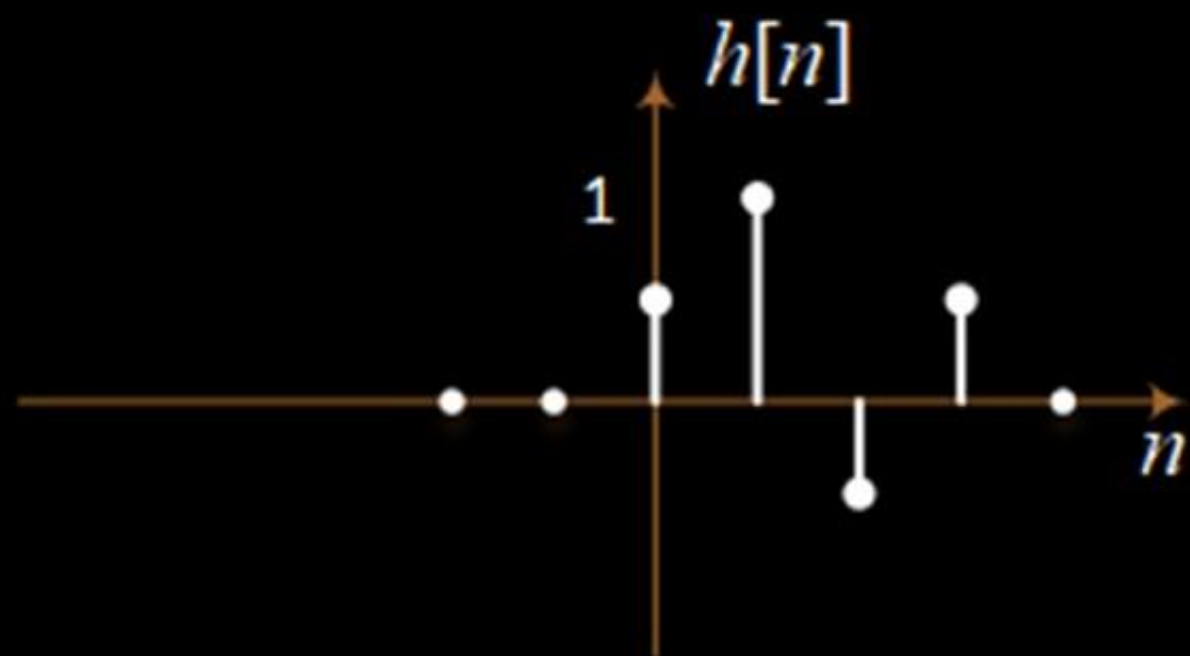
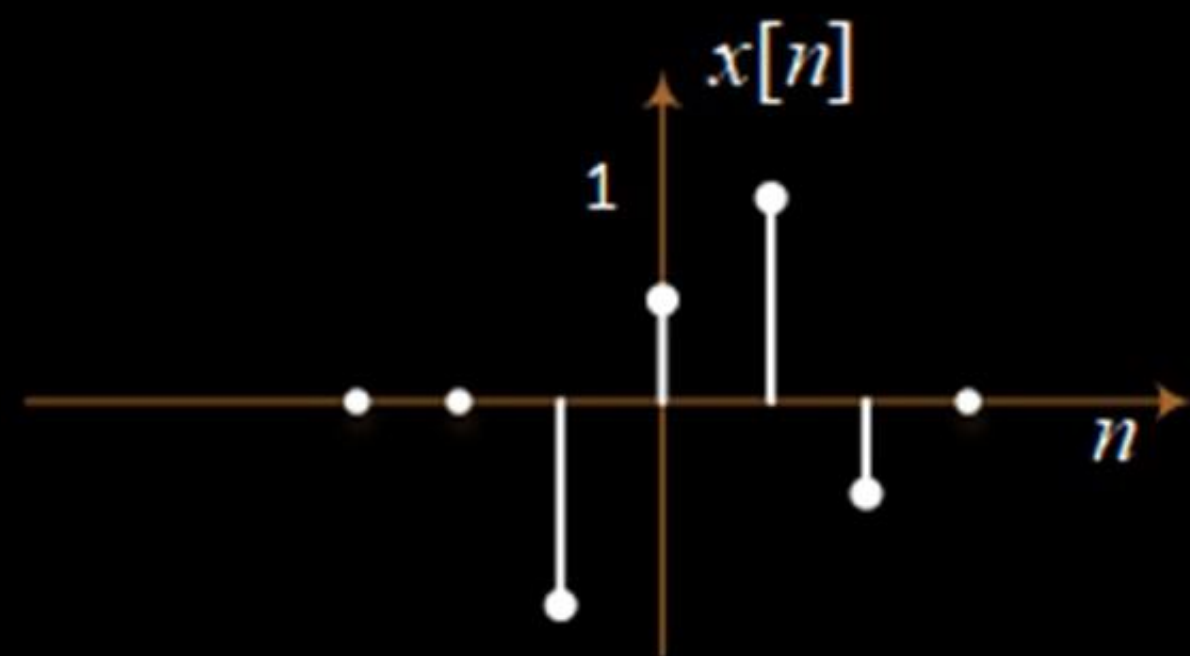
For  $n = +3$



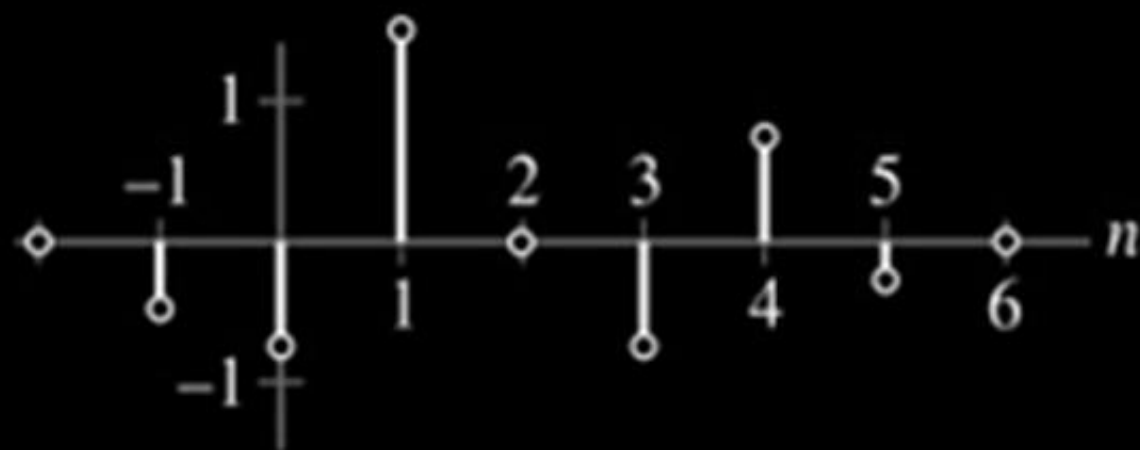
For  $n = +3$



$$x[k]h[n-k] = \frac{1}{4} - \frac{1}{2} - \frac{1}{2} = -0.75$$

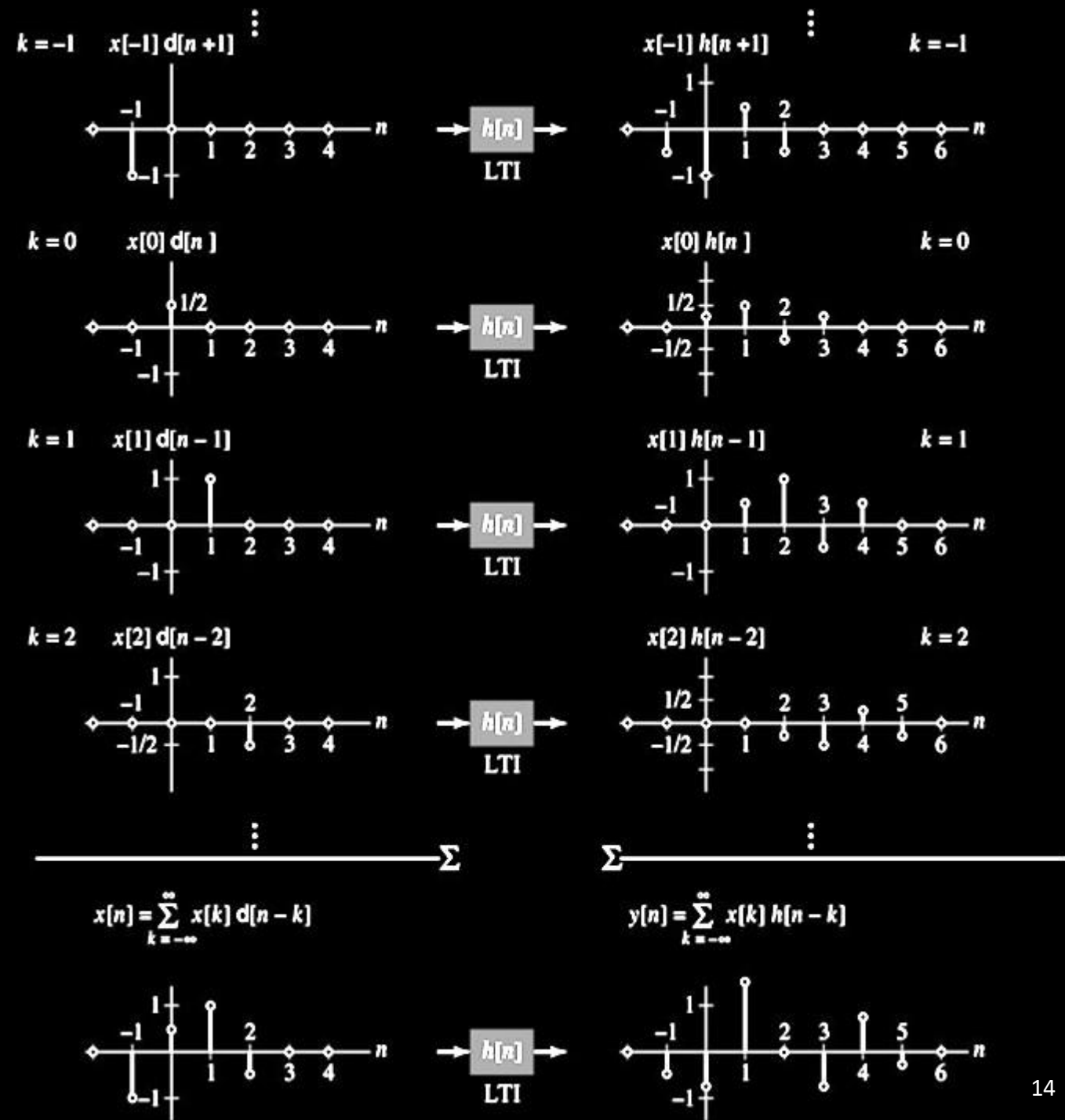


$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$



**Figure 2.2b**  
**(p. 101)**

(b) The decomposition of the input  $x[n]$  into a weighted sum of time-shifted impulses results in an output  $y[n]$  given by a weighted sum of time-shifted impulse responses.





**EXAMPLE 2.1 MULTIPATH COMMUNICATION CHANNEL: DIRECT EVALUATION OF THE CONVOLUTION SUM** Consider the discrete-time LTI system model representing a two-path propagation channel described in Section 1.10. If the strength of the indirect path is  $a = 1/2$ , then

$$y[n] = x[n] + \frac{1}{2}x[n-1].$$

Letting  $x[n] = \delta[n]$ , we find that the impulse response is

$$h[n] = \begin{cases} 1, & n = 0 \\ \frac{1}{2}, & n = 1 \\ 0, & \text{otherwise} \end{cases}.$$

Determine the output of this system in response to the input

$$x[n] = \begin{cases} 2, & n = 0 \\ 4, & n = 1 \\ -2, & n = 2 \\ 0, & \text{otherwise} \end{cases}.$$

**Solution:** First, write  $x[n]$  as the weighted sum of time-shifted impulses:

$$x[n] = 2\delta[n] + 4\delta[n - 1] - 2\delta[n - 2].$$

Here, the input is decomposed as a weighted sum of three time-shifted impulses because the input is zero for  $n < 0$  and  $n > 2$ . Since a weighted, time-shifted impulse input,  $\gamma\delta[n - k]$ , results in a weighted, time-shifted impulse response output,  $\gamma h[n - k]$ , Eq. (2.4) indicates that the system output may be written as

$$y[n] = 2h[n] + 4h[n - 1] - 2h[n - 2].$$

Summing the weighted and shifted impulse responses over  $k$  gives

$$y[n] = \begin{cases} 0, & n < 0 \\ 2, & n = 0 \\ 5, & n = 1 \\ 0, & n = 2 \\ -1, & n = 3 \\ 0, & n \geq 4 \end{cases}.$$

## 2.3 Convolution Sum Evaluation Procedure

- When the input is of long duration, recall that the convolution sum is expressed as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

- Defining

$$w_n[k] = x[k] h[n-k] \quad (2.5)$$

- Hence,

$$y[n] = \sum_{k=-\infty}^{\infty} w_n[k] \quad (2.6)$$

**EXAMPLE 2.2 CONVOLUTION SUM EVALUATION BY USING AN INTERMEDIATE SIGNAL**  
Consider a system with impulse response

$$h[n] = \left(\frac{3}{4}\right)^n u[n].$$

Use Eq. (2.6) to determine the output of the system at times  $n = -5$ ,  $n = 5$ , and  $n = 10$  when the input is  $x[n] = u[n]$ .

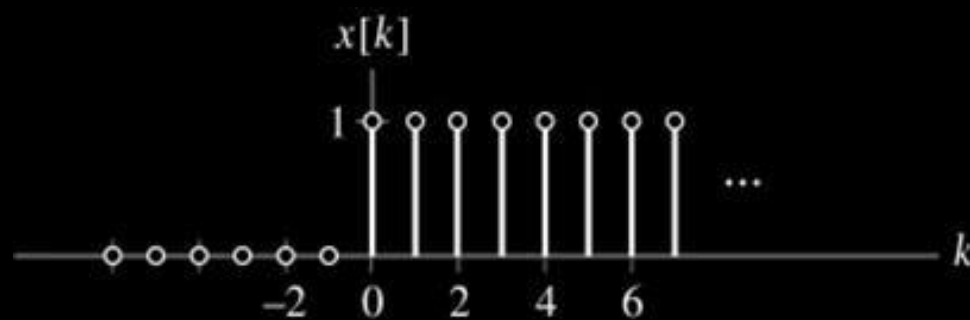
**Solution:** Here, the impulse response and input are of infinite duration, so the procedure followed in Example 2.1 would require summing a large number of time-shifted impulse responses to determine  $y[n]$  for each  $n$ . By using Eq. (2.6), we form only one signal,  $w_n[k]$ , for each  $n$  of interest. Figure 2.3(a) depicts  $x[k]$  superimposed on the reflected and time-shifted impulse response  $h[n - k]$ . We see that

$$h[n - k] = \begin{cases} \left(\frac{3}{4}\right)^{n-k}, & k \leq n \\ 0, & \text{otherwise} \end{cases}.$$

**Figure 2.3 (p. 103)**

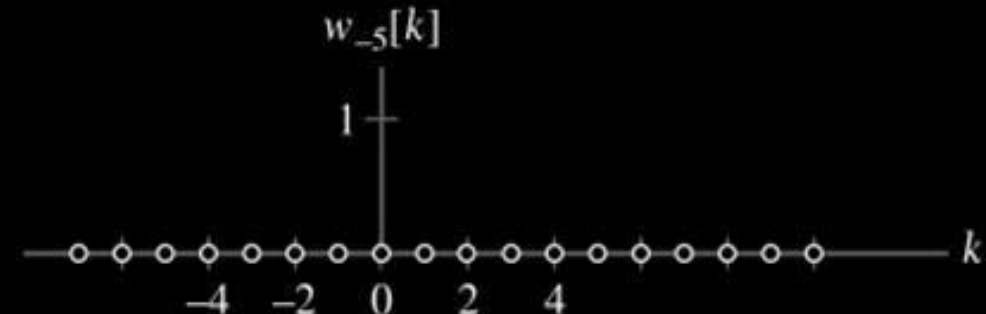
Evaluation of Eq. (2.6) in Ex. 2.2.

a) The input signal  $x[k]$  above the reflected and time-shifted impulse response  $h[n-k]$ , depicted as a function of  $k$ .



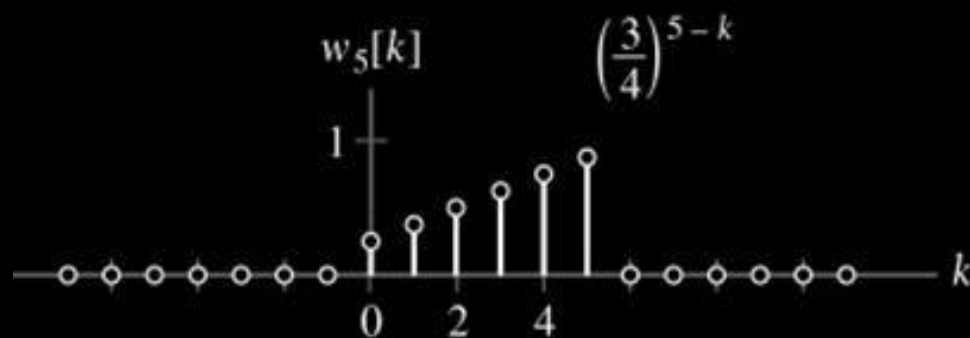
(a)

b) The product signal  $w_5[k]$  used to evaluate  $y[-5]$ .



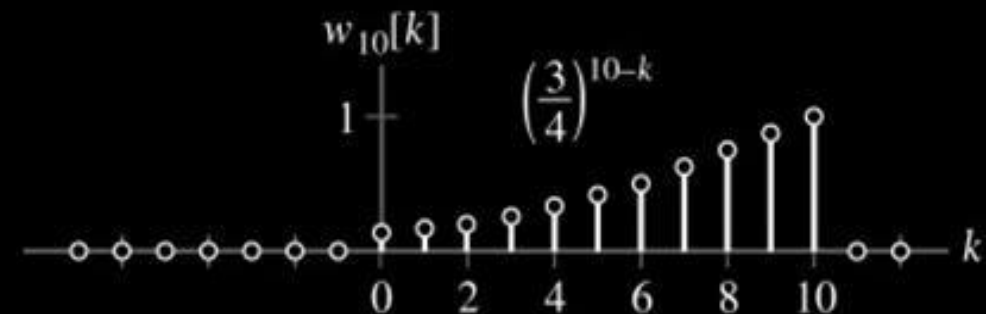
(b)

c) The product signal  $w_5[k]$  used to evaluate  $y[5]$ .



(c)

d) The product signal  $w_{10}[k]$  used to evaluate  $y[10]$ .



(d)

The intermediate signal  $w_n[k]$  is now easily obtained by means of Eq. (2.5). Figures 2.3(b), (c), and (d) depict  $w_n[k]$  for  $n = -5$ ,  $n = 5$ , and  $n = 10$ , respectively. We have

$$w_{-5}[k] = 0,$$

and thus Eq. (2.6) gives  $y[-5] = 0$ . For  $n = 5$ , we have

$$w_5[k] = \begin{cases} \left(\frac{3}{4}\right)^{5-k}, & 0 \leq k \leq 5 \\ 0, & \text{otherwise} \end{cases},$$

so Eq. (2.6) gives

$$y[5] = \sum_{k=0}^5 \left(\frac{3}{4}\right)^{5-k},$$

which represents the sum of the nonzero values of the intermediate signal  $w_5[k]$  shown in Fig. 2.3(c). We then factor  $\left(\frac{3}{4}\right)^5$  from the sum and apply the formula for the sum of a finite geometric series (see Appendix A.3) to obtain

$$\begin{aligned} y[5] &= \left(\frac{3}{4}\right)^5 \sum_{k=0}^5 \left(\frac{4}{3}\right)^k \\ &= \left(\frac{3}{4}\right)^5 \frac{1 - \left(\frac{4}{3}\right)^6}{1 - \left(\frac{4}{3}\right)} = 3.288. \end{aligned}$$

## HATIRLATMA!

$r \neq 1$  olmak koşuluyla bir geometrik serinin ilk  $n$  terimi toplamı

$$a + ar + ar^2 + ar^3 + \dots + ar^{n-1} = \sum_{k=0}^{n-1} ar^k = a \frac{1 - r^n}{1 - r}$$

$n$  sonsuza giderken

$$s = \sum_{k=0}^{\infty} ar^k = \frac{a}{1 - r}$$

biçimini alır.  $a = 1$  ise bu ifade

$$1 + r + r^2 + r^3 + \dots = \frac{1}{1 - r}$$

Last, for  $n = 10$ , we see that

$$w_{10}[k] = \begin{cases} \left(\frac{3}{4}\right)^{10-k}, & 0 \leq k \leq 10 \\ 0, & \text{otherwise} \end{cases},$$

and Eq. (2.6) gives

$$\begin{aligned} y[10] &= \sum_{k=0}^{10} \left(\frac{3}{4}\right)^{10-k} \\ &= \left(\frac{3}{4}\right)^{10} \sum_{k=0}^{10} \left(\frac{4}{3}\right)^k \\ &= \left(\frac{3}{4}\right)^{10} \frac{1 - \left(\frac{4}{3}\right)^{11}}{1 - \left(\frac{4}{3}\right)} = 3.831. \end{aligned}$$

Note that in this example  $w_n[k]$  has only two different mathematical representations. For  $n < 0$ , we have  $w_n[k] = 0$ , since there is no overlap between the nonzero portions of  $x[k]$  and  $h[n - k]$ . When  $n \geq 0$ , the nonzero portions of  $x[k]$  and  $h[n - k]$  overlap on the interval  $0 \leq k \leq n$ , and we may write

$$w_n[k] = \begin{cases} \left(\frac{3}{4}\right)^{n-k}, & 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}.$$

Hence, we may determine the output for an arbitrary  $n$  by using the appropriate mathematical representation for  $w_n[k]$  in Eq. (2.6). ■